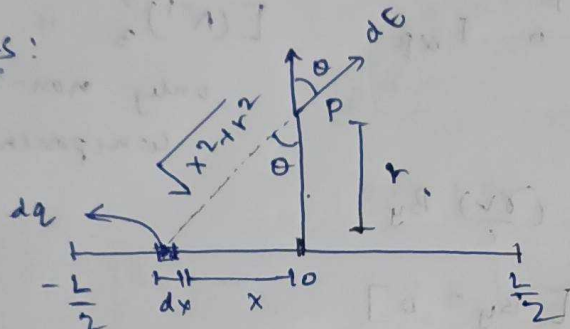


- ① Find the EF @ a distance r away from the midpoint of a straight line segment of length L , which carries a line charge density λ . For the case where $r \ll L$, express the EF at the point r . Does E depend on length L ?

Ans:



$$dq = \lambda dx$$

$$dE = \frac{dq}{4\pi\epsilon_0 (x^2 + r^2)} = \frac{\lambda dx}{4\pi\epsilon_0 (x^2 + r^2)}$$

dE has 2 components — one $\parallel$ to the wire & one $\perp$ to the wire. The $\perp$ components from an equal segment present at the opposite side cancel each other, while the $\parallel$ components add up.

$$\therefore dE_{net} = 2 dE \cos \theta$$

$$= \frac{\lambda dx}{2\pi\epsilon_0 (x^2 + r^2)} \cdot \frac{r}{(x^2 + r^2)^{1/2}}$$

$$\rightarrow dE_{net} = \frac{\lambda r dx}{2\pi\epsilon_0 (x^2 + r^2)^{3/2}}$$

$$\rightarrow \int E_{net} = \frac{\lambda r}{2\pi\epsilon_0} \int_0^{L/2} \frac{dx}{(x^2 + r^2)^{3/2}}$$

$$\text{Let } x = r \tan \theta \rightarrow dx = r \sec^2 \theta d\theta$$

$$E_{net} = \frac{\lambda r}{2\pi\epsilon_0} \int_0^{\tan^{-1}(L/2r)} \frac{r \cdot \sec^2\theta \cdot d\theta}{r^3 \sec^3\theta} = \frac{\lambda}{2\pi\epsilon_0 r} \int_0^{\tan^{-1}(L/2r)} \cos\theta \cdot d\theta$$

$$= \frac{\lambda}{2\pi\epsilon_0 r} \sin\theta \Big|_0^{\tan^{-1}(L/2r)}$$

$$E_{net} = \frac{\lambda}{2\pi\epsilon_0 r} \frac{L}{\sqrt{L^2 + 4r^2}}$$

$$\boxed{E_{net} = \frac{\lambda}{2\pi\epsilon_0 r} \frac{L}{\sqrt{L^2 + 4r^2}} \hat{k}} \quad (\text{Ans})$$

② $r \ll L$:

$$L^2 + 4r^2 \approx L^2$$

$$\Rightarrow \sqrt{L^2 + 4r^2} \approx L$$

$$E_{net} = \frac{\lambda}{2\pi\epsilon_0 r} \frac{L}{L} = \boxed{\frac{\lambda}{2\pi\epsilon_0 r} = E_{net}} \quad (\text{Ans})$$

E does not depend on length L .

- ② Find the solution of the Laplace's Equation $\nabla^2 V = 0$ in 1 dimension. Show that the potential $V(x)$ at any point ~~does not~~ can be expressed as the average of its right & left values of the potential. Does the potential admit any local extremum?

Ans :

$$\nabla^2 V = 0$$

$$\rightarrow \nabla V = A$$

$$\rightarrow \boxed{V(x) = Ax + B} \quad (\text{Ans})$$

$$V(x+a) = A(x+a) + B \quad \text{--- (I)}$$

$$V(x-a) = A(x-a) + B \quad \text{--- (II)}$$

$$\therefore v(x+a) + v(x-a) = 2(Ax+B) = 2v(x)$$

$$\rightarrow \boxed{v(x) = \frac{1}{2} [v(x+a) + v(x-a)]} \quad (\text{Proved})$$

$\therefore v(x)$ is the avg. of its values on its left & right, hence it does not admit any local extremum. This is because at a local extremum, the minimum / maximum value is not the average of its surrounding values.